Blackmon Lab

Evolutionary Biology

Environmental Instability and Chromosome Evolution in African Killifish

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Background

Annual killifish experience extreme population bottlenecks driven by seasonal rainfall patterns in their native African habitats. The puddles they live in appear during the rainy season and then completely dry up. This project investigates whether precipitation seasonality, used as a proxy for effective population size, predicts chromosomal evolution in the family *Nothobranchiidae*. Higher seasonality is expected to result in shorter, more variable wet periods, leading to smaller and more unstable populations. We hypothesize that high seasonality drives faster chromosomal divergence, as frequent bottlenecks enhance genetic drift and reduce the efficiency of natural selection, allowing chromosomal changes to accumulate. A comparative analysis will be used to test whether variation in seasonality drives differences in rates of chromosome evolution.

Epiplatys bifasciatus, Mody, K.

Aphyosemion volcanum, Terceira, A.C.

Epiplatys lamottei, Nilsson, K.

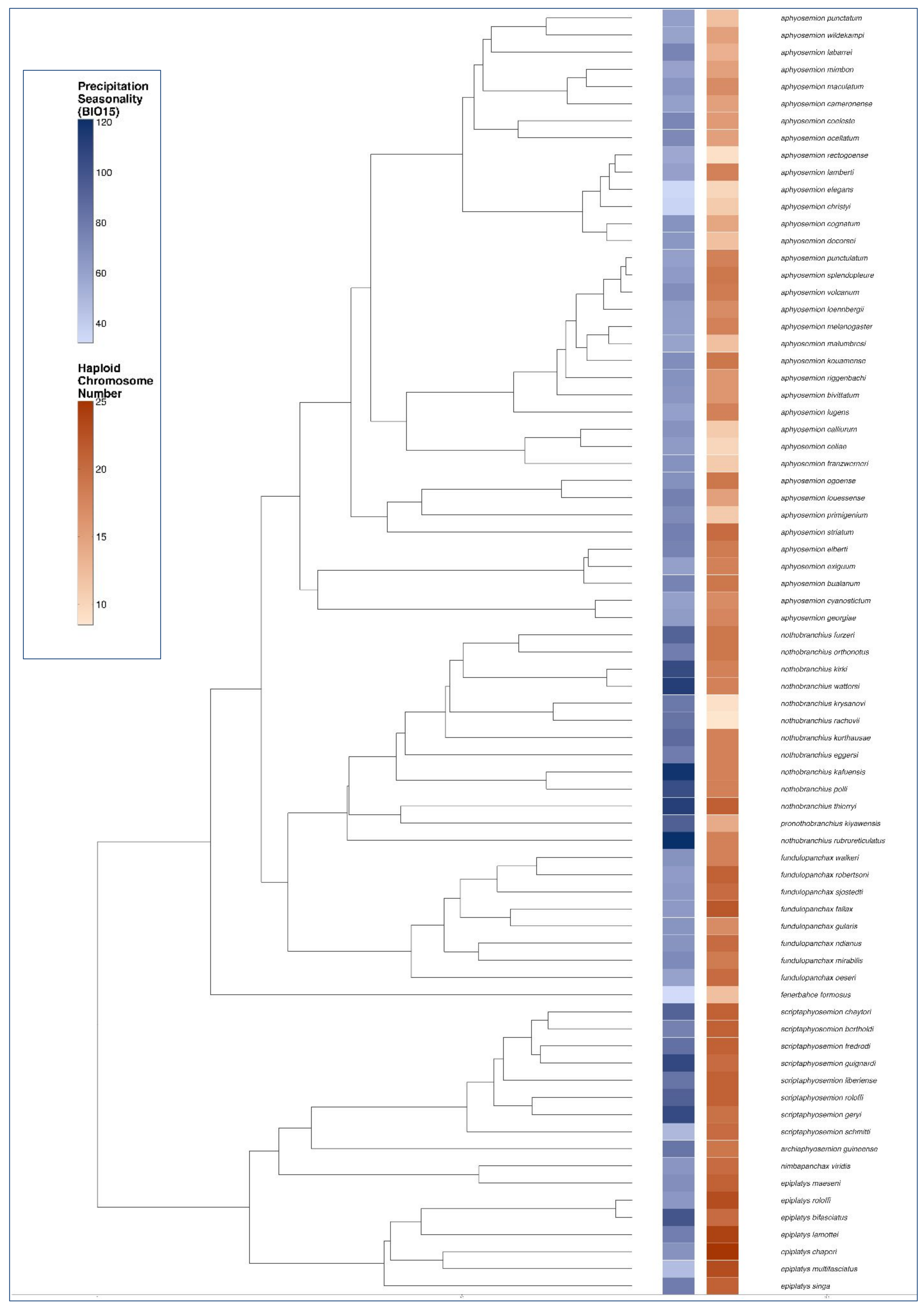
Aphyosemion labarrei, Van der Zee, J.R.

Methods & AI Usage

Chromosome evolution was modeled using ChromePlus in R, which estimates rates of fission and fusion along a phylogeny. Seasonality was coded as a binary trait (high/low) based on WorldClim BIO15 data. Geographic data was collected via GBIF (Global Biodiversity Information Facility). Transition rates between seasonality states were estimated using ChromePlus. The phylogeny was provided by the Blackmon Lab. MCMC chains were run for 10000 generations with 25% burn-in. Generative AI (Claude/Gemini/ChatGPT) was used in this project assisting in code development, data visualization, and analytical development. All outputs were reviewed by authors.

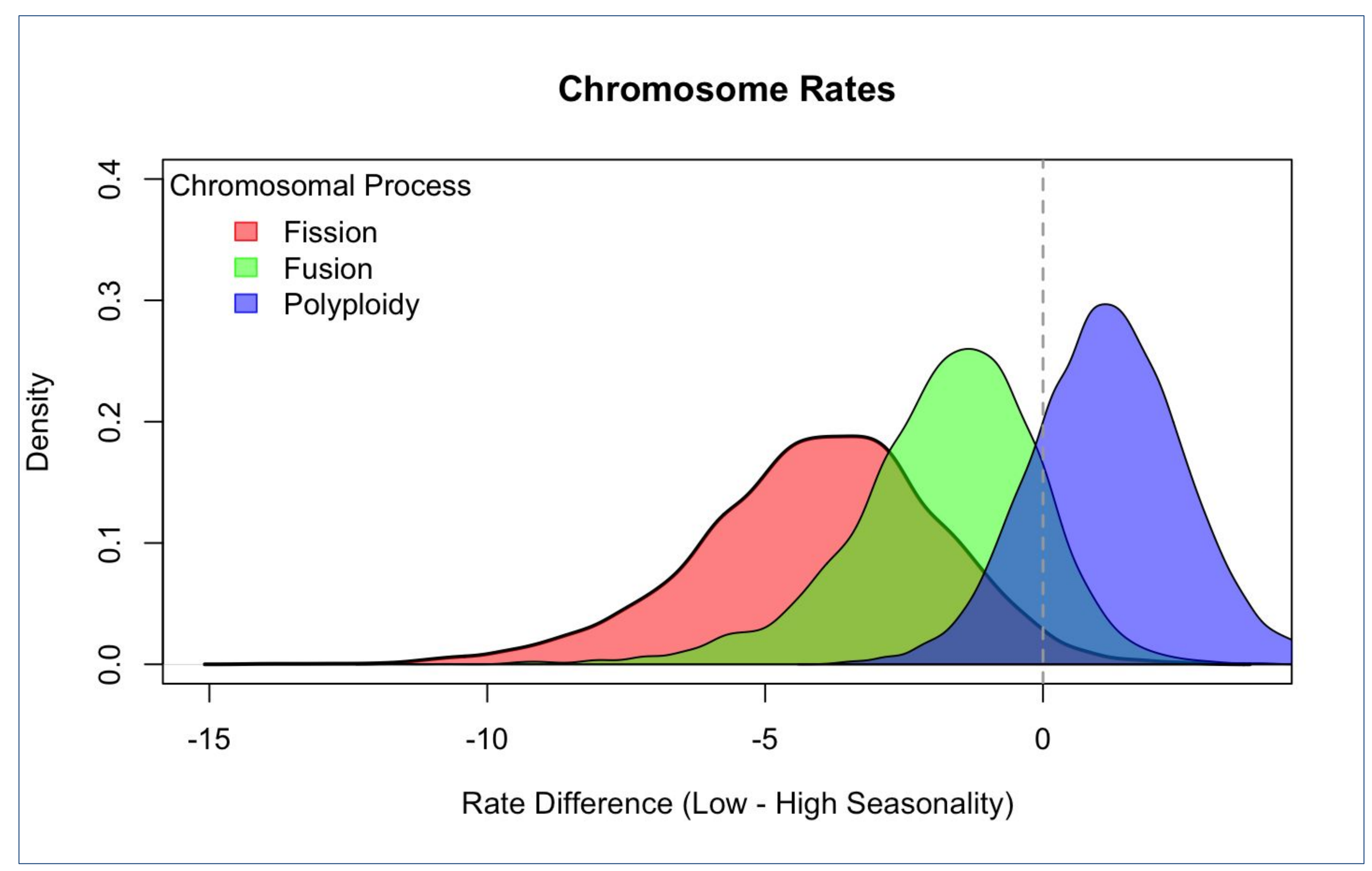
Results

Phenotype & Species Summary: After pruning of the original phylogeny provided by the Blackmon Lab, chromosome numbers were matched to species that had geographical data in the GBIF. This resulted in chromosome numbers ranging from 8 to 25 across a total of 155 species. The seasonality data from GBIF was treated as a binary trait (high vs. low) to accommodate to the ChromePlus model requirements, despite seasonality being a typically continuous.



Rate Analysis: Dysploidy rates (fission and fusion) were substantially elevated in high seasonality areas. The polyploidy rates are not impacted by seasoanlity as polyploidy is typically guided by different evolutionary mechanisms.

Statistical Analysis: We found that *Nothobranchiidae* species inhabiting high precipitation seasonality environments (Mean BIO15 ≥ median threshold) characterized by greater climatic fluctuations and more frequent population bottlenecks exhibited slightly higher median chromosome numbers than low-seasonality species, though this non-phylogenetic comparison was only marginally significant (Wilcoxon $p = 0.056$). Phylogenetic ChromePlus modeling with 10,000 MCMC iterations revealed substantially elevated rates of chromosomal rearrangement in high-seasonality lineages. The median fission rate in high-seasonality species (5.81) was 3.25× higher than in low-seasonality species (1.75), with a rate ratio 95% credible interval of [1.07, 14.54] that excludes 1.0, and a posterior probability of 0.98. Similarly, the median fusion rate was 2.59× higher in high-seasonality species (2.74 vs. 1.05), though this difference was less conclusive (rate ratio 95% CI [0.45, 21.74]; PP = 0.87). These results are consistent with the hypothesis that seasonal population bottlenecks reduce effective population size, weakening purifying selection against underdominant chromosomal rearrangements and facilitating their fixation via genetic drift. Examples of species supporting this pattern include *Nothobranchius krysanovi* (BIO15 = 80.68, n = 9) and *N. rachovii* (BIO15 = 83.62, n = 8–9), which inhabit highly seasonal habitats and have undergone dramatic chromosomal reductions from the conserved n = 18, while low-seasonality *Nothobranchius* such as *N. robustus* (BIO15 = 46.78) and *N. ugandensis* (BIO15 = 44.07) retain the ancestral chromosome number. It is important to note that the binary coding of precipitation seasonality via a median split is a simplification of a continuous variable, and the credible intervals for fusion remain wide, so further research with larger taxon sampling and continuous-trait models could yield more conclusive results.



Discussion

Findings partially support the hypothesis that environmental instability accelerates chromosomal evolution. Elevated fission and fusion rates in high-seasonality lineages are consistent with enhanced genetic drift in smaller, more bottlenecked populations. Polyploidy rates did not differ by seasonality, likely because whole-genome duplication arises through mechanisms (e.g., unreduced gamete formation) less influenced by effective population size. Most high seasonality species is of the genus, *Nothobranchius,* which could partially reflect claude-specieifc effects rather than environment. Ancestral state reconstruction estimated both 0→1 and 1→0 transitions in seasonality, indicating bidirectional evolutionary shifts rather than a unidirectional trend. Marginally significant difference in chromosome number ($p = 0.056$) suggests a potential relationship warranting further investigation with expanded sampling.

Conclusions & Acknowledgements

High precipitation seasonality used as a proxy for effective population size is associated with elevated dysploidy rates in African killifish, supporting the hypothesis that population bottlenecks facilitate chromosomal change. These results indicate that effective population size may play a pivotal role in chromosome number evolution. Polyploidy appears mechanistically distinct and unaffected by seasonality. Future work should incorporate continuous seasonality metrics and broader taxonomic sampling.